

# Small-Strain Flexoelectric Pure Bending with Generalized Traction

## 1 Setting

Consider a beam in small-strain pure bending. The reference thickness coordinate is

$$Y \in [-a, a], \quad a = \frac{H}{2}.$$

The imposed axial bending strain is taken as

$$\varepsilon_{11}(Y) = -\kappa Y,$$

and the transverse strain is left unknown:

$$\varepsilon_{22}(Y) = v(Y).$$

This is the important difference from the simpler plane-stress-style derivation: the local transverse stress is not prescribed to be zero pointwise. Instead, the free surface condition is imposed on the generalized traction.

For an isotropic linear elastic solid,

$$C_L = \lambda + 2\mu, \quad C_T = \lambda,$$

so the local transverse stress is

$$\sigma_{22} = C_L v - C_T \kappa Y. \tag{1}$$

## 2 Open-Circuit Electric Field

Let a prime denote differentiation with respect to  $Y$ . The one-dimensional open-circuit electric displacement relation is written as

$$D_2 = \epsilon E_2 + \mu_L v' - \mu_T \kappa. \tag{2}$$

For the open-circuit through-thickness response,

$$D_2 = 0,$$

and therefore

$$E_2(Y) = \frac{\mu_T \kappa - \mu_L v'(Y)}{\epsilon}. \tag{3}$$

### 3 Higher-Order Stress After Elimination

Before eliminating the electric field, the relevant higher-order stress component is

$$S_{222} = H_L v' - H_T \kappa - \mu_L E_2. \quad (4)$$

Using Eq. (3),

$$S_{222} = G_L v' - G_T \kappa, \quad (5)$$

where

$$G_L = H_L + \frac{\mu_L^2}{\epsilon}, \quad G_T = H_T + \frac{\mu_L \mu_T}{\epsilon}. \quad (6)$$

The coefficients  $H_L, H_T$  depend on the strain-gradient/flexoelectric material form used in the numerical model. Two useful cases are:

| model       | $H_L, H_T$                                                                                                   | $G_L, G_T$                                                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| bare        | $C_L \ell^2, C_T \ell^2$                                                                                     | $C_L \ell^2 + \mu_L^2/\epsilon, C_T \ell^2 + \mu_L \mu_T/\epsilon$                                                                                     |
| transformed | $C_L \ell^2 - \frac{\mu_L^2}{\epsilon - \epsilon_0}, C_T \ell^2 - \frac{\mu_L \mu_T}{\epsilon - \epsilon_0}$ | $C_L \ell^2 - \frac{\epsilon_0 \mu_L^2}{\epsilon(\epsilon - \epsilon_0)}, C_T \ell^2 - \frac{\epsilon_0 \mu_L \mu_T}{\epsilon(\epsilon - \epsilon_0)}$ |

The plotting script uses the bare model by default. This matches a numerical run where the additional  $\bar{H}$  flexoelectric correction is not subtracted. The transformed model is kept as an option for runs where that correction is included.

### 4 Generalized Free-Surface Conditions

The through-thickness equilibrium and ordinary traction condition are

$$(\sigma_{22} - S'_{222})' = 0, \quad (\sigma_{22} - S'_{222})|_{Y=\pm a} = 0.$$

Thus,

$$\sigma_{22} - S'_{222} = 0. \quad (7)$$

Using Eqs. (1) and (5), this becomes

$$C_L v - C_T \kappa Y - G_L v'' = 0. \quad (8)$$

Equivalently,

$$v - \xi^2 v'' = \beta \kappa Y, \quad \xi = \sqrt{\frac{G_L}{C_L}}, \quad \beta = \frac{C_T}{C_L}. \quad (9)$$

The higher-order traction is also free at the top and bottom surfaces:

$$S_{222}(\pm a) = 0.$$

Therefore,

$$v'(\pm a) = \gamma \kappa, \quad \gamma = \frac{G_T}{G_L}. \quad (10)$$

Equations (9) and (10) are the replacement for the pointwise assumption  $\sigma_{22} = 0$ .

## 5 Closed Form

Define

$$\eta = \gamma - \beta = \frac{G_T}{G_L} - \frac{C_T}{C_L}.$$

The solution of Eq. (9) satisfying Eq. (10) is

$$v(Y) = \beta\kappa Y + \kappa\xi\eta \frac{\sinh(Y/\xi)}{\cosh(a/\xi)}, \quad (11)$$

and

$$v'(Y) = \beta\kappa + \kappa\eta \frac{\cosh(Y/\xi)}{\cosh(a/\xi)}. \quad (12)$$

The local transverse stress is then

$$\sigma_{22} = C_L\kappa\xi\eta \frac{\sinh(Y/\xi)}{\cosh(a/\xi)}.$$

It is generally not zero. However, this is exactly balanced by  $S'_{222}$ , so the generalized traction  $\sigma_{22} - S'_{222}$  vanishes.

## 6 Electric Potential

Substitution of Eq. (12) into Eq. (3) gives

$$E_2(Y) = \frac{\kappa}{\epsilon} \left[ \mu_T - \mu_L\beta - \mu_L\eta \frac{\cosh(Y/\xi)}{\cosh(a/\xi)} \right]. \quad (13)$$

Let

$$A_0 = \frac{\kappa}{\epsilon} (\mu_T - \mu_L\beta), \quad B_0 = \frac{\mu_L\kappa\xi\eta}{\epsilon}.$$

Since  $E_2 = -\phi'$ , and choosing the bottom surface gauge

$$\phi(-a) = 0,$$

the electric potential is

$$\boxed{\phi(Y) = -A_0(Y + a) + B_0 \left[ \frac{\sinh(Y/\xi)}{\cosh(a/\xi)} + \tanh(a/\xi) \right]}. \quad (14)$$

When  $G_L = 0$ , the boundary-layer length  $\xi$  disappears and the reduced local solution is

$$v(Y) = \beta\kappa Y, \quad E_2 = A_0, \quad \phi(Y) = -A_0(Y + a).$$

If  $G_L < 0$ , the boundary-layer length is not real, so this closed form is not an admissible real-valued solution.

## 7 Implementation Notes

The plotting script implements Eq. (14). It reads  $H, L, E, \nu, \ell, \epsilon, \mu_L, \mu_T$  from `log.txt`, extracts  $\kappa$  from the selected ParaView step, and uses the bottom-surface gauge  $\phi(-H/2) = 0$ . Its default analytical model is `bare`; `transformed` and `local` are available through `--analytical-model`.